Appendix for "Estimating Subgraph Importance with Structural Prior Domain Knowledge"

Contents

This appendix provides additional details, experimental results, and extended analyses that supplement the main paper. In particular, we include dataset statistics, model architecture descriptions, baseline configurations, and full fidelity scores used in our evaluation.

A Datasets

The statistics of the datasets for the experiment are in Table 1. The `Alkane-Carbonyl`, `FluorideCarbonyl`, and `Benzene` datasets are obtained from the *GraphXAI* library[1]. The `UPFD`, `BA2-Motif`, and `BAMultiShapes` datasets are accessed through the *PyTorch Geometric* library. Additionally, we use `bace`, `bbbp` datasets from the *OGB* benchmark suite. `Ames` is retrieved from the *Therapeutics Data Commons (TDC)*.

B Model Architecture

We implemented a graph classification model with the following components. For datasets with continuous node features (e.g., social networks, biological, and synthetic data), we apply an MLP projection for continuous features and an embedding layer for molecular atom indices. And then 3 stacks of message-passing GNN layers (e.g., GCN, GIN, GAT, and GraphSAGE) are used to encode node representations. These node embeddings are aggregated into a graph-level representation using a global pooling operation, selected from mean, sum, or max pooling. Our method is not restricted to these pooling strategies and can be applied to any readout function, as long as the output dimensions between node-level and graph-level representations are compatible. Finally, the pooled graph representation is passed through a multi-layer perceptron (MLP) for classification.

C Baseline Methods

We implemented GNNExplainer, PGExplainer, GraphMask, Saliency, and GuidedBackprop via the *torch-geometric* library. For PGExplainer's edge-level scores, node importance was obtained by summing its incoming edge scores. SubgraphX uses publicly available code[2]. To evaluate the importance by Subgraphx, we predefined the number of important nodes to be selected from each explanation. For each method, we used the hyperparameters as specified in their original papers. For methods that allow hyperparameter tuning, we additionally adjusted their settings to improve performance. Hyperparameters for tunable methods are listed in Table 2.

[1] `https://github.com/mims-harvard/GraphXAI`
[2] `https://www.kaggle.com/code/yannikmahlau/subgraphx`

Table 1: Summary of datasets used in this study.

Dataset	#Graphs	Avg. #Nodes	Avg. #Edges	#Tasks
IMDB	1,000	19.79	193.25	2
ENZYMES	600	32.63	62.14	6
PROTEINS	1,113	39.06	72.82	2
UPFD	45,000	25.0	24.0	2
MULTI	1,000	40.0	80.0	2
MOTIF	1,000	25.0	50.0	2
MUTAG	188	17.9	19.8	2
Benzene	12,000	20.58	43.65	2
Fluoride	8,671	21.36	44.95	2
Alkane	4,326	21.13	45.37	2
bace	1,513	34.1	36.9	2
Ames	2,930	20.0	22.0	2
bbbp	2,039	24.1	26.0	2

Table 2: Hyperparameter settings of baseline methods.

Method	Parameter	Setting
eXEL	λ	{0.001, 0.01, 0.1, 1, 10, 100}
PGExplainer	Learning rate	{0.001,0.002,0.003}
GNNExplainer	Learning rate	{0.001, 0.01, 0.1}
GraphMask	Learning rate	{0.01,0.02,0.03}
SubgraphX	Shapley samples	100
	MCTS iterations	20
	λ (UCB weight)	{5, 10}

D Detailed Fidelity Scores

In the main paper, we report fidelity scores averaged over all readout functions for a unified comparison. In this appendix, we additionally provide the full set of fidelity values without averaging, fidelity scores computed separately for each readout function. All experimental results are presented in the tables below. For ease of comparison, the best-performing method in each row is shown in **bold**, while the runner-up is marked with an underline.

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.125_{\pm 0.134}$	$\mathbf{0.290_{\pm 0.265}}$	$0.005_{\pm 0.024}$	$0.071_{\pm 0.073}$	$0.015_{\pm 0.038}$	$0.043_{\pm 0.072}$	$0.120_{\pm 0.174}$	$0.031_{\pm 0.041}$	$\underline{0.211_{\pm 0.194}}$
alkane	$0.024_{\pm 0.016}$	$\mathbf{0.515_{\pm 0.351}}$	$0.019_{\pm 0.027}$	$\underline{0.164_{\pm 0.114}}$	$0.150_{\pm 0.121}$	$0.107_{\pm 0.068}$	$0.063_{\pm 0.061}$	$0.084_{\pm 0.149}$	$0.008_{\pm 0.012}$
ames	$\underline{0.148_{\pm 0.038}}$	$\mathbf{0.187_{\pm 0.065}}$	$0.016_{\pm 0.032}$	$0.110_{\pm 0.028}$	$0.047_{\pm 0.030}$	$0.064_{\pm 0.037}$	$0.091_{\pm 0.048}$	$0.103_{\pm 0.055}$	$0.131_{\pm 0.053}$
bace	$\mathbf{0.129_{\pm 0.097}}$	$0.076_{\pm 0.018}$	$0.007_{\pm 0.043}$	$0.074_{\pm 0.042}$	$0.040_{\pm 0.022}$	$0.022_{\pm 0.042}$	$\underline{0.089_{\pm 0.051}}$	$0.067_{\pm 0.077}$	$0.052_{\pm 0.040}$
bbbp	$\underline{0.188_{\pm 0.097}}$	$\mathbf{0.230_{\pm 0.066}}$	$0.067_{\pm 0.049}$	$0.161_{\pm 0.039}$	$0.052_{\pm 0.027}$	$0.084_{\pm 0.019}$	$0.169_{\pm 0.113}$	$0.067_{\pm 0.023}$	$0.132_{\pm 0.064}$
benzene	$\mathbf{0.450_{\pm 0.052}}$	$\underline{0.403_{\pm 0.100}}$	$0.029_{\pm 0.013}$	$0.152_{\pm 0.070}$	$0.092_{\pm 0.031}$	$0.080_{\pm 0.010}$	$0.219_{\pm 0.103}$	$0.283_{\pm 0.117}$	$0.344_{\pm 0.046}$
fluoride	$0.342_{\pm 0.138}$	$\mathbf{0.707_{\pm 0.119}}$	$0.384_{\pm 0.110}$	$0.502_{\pm 0.113}$	$0.612_{\pm 0.140}$	$0.266_{\pm 0.135}$	$0.403_{\pm 0.204}$	$\underline{0.645_{\pm 0.078}}$	$0.272_{\pm 0.082}$

Table 3: Fidelity^{+} scores computed using the Mean method (higher is better) with GCN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.184_{\pm 0.132}$	$\mathbf{0.338_{\pm 0.180}}$	$-0.024_{\pm 0.030}$	$0.071_{\pm 0.078}$	$0.114_{\pm 0.146}$	$0.012_{\pm 0.064}$	$0.008_{\pm 0.055}$	$-0.003_{\pm 0.043}$	$\underline{0.223_{\pm 0.158}}$
alkane	$\underline{0.350_{\pm 0.107}}$	$\mathbf{0.352_{\pm 0.106}}$	$0.005_{\pm 0.026}$	$0.039_{\pm 0.031}$	$0.175_{\pm 0.190}$	$0.018_{\pm 0.036}$	$0.010_{\pm 0.010}$	$0.003_{\pm 0.011}$	$0.004_{\pm 0.017}$
ames	$\mathbf{0.178_{\pm 0.062}}$	$0.155_{\pm 0.047}$	$0.034_{\pm 0.027}$	$0.098_{\pm 0.025}$	$0.055_{\pm 0.027}$	$0.062_{\pm 0.021}$	$0.076_{\pm 0.010}$	$0.068_{\pm 0.030}$	$\underline{0.162_{\pm 0.086}}$
bace	$\underline{0.242_{\pm 0.089}}$	$\mathbf{0.259_{\pm 0.046}}$	$0.081_{\pm 0.043}$	$0.176_{\pm 0.083}$	$0.145_{\pm 0.053}$	$0.092_{\pm 0.031}$	$0.102_{\pm 0.042}$	$0.168_{\pm 0.048}$	$0.150_{\pm 0.036}$
bbbp	$\mathbf{0.079_{\pm 0.046}}$	$0.062_{\pm 0.026}$	$0.007_{\pm 0.013}$	$\underline{0.072_{\pm 0.040}}$	$0.010_{\pm 0.011}$	$0.035_{\pm 0.028}$	$0.042_{\pm 0.033}$	$0.016_{\pm 0.015}$	$0.066_{\pm 0.026}$
benzene	$0.453_{\pm 0.058}$	$\mathbf{0.520_{\pm 0.078}}$	$0.132_{\pm 0.059}$	$0.157_{\pm 0.071}$	$\underline{0.497_{\pm 0.043}}$	$0.058_{\pm 0.023}$	$0.134_{\pm 0.126}$	$0.460_{\pm 0.041}$	$0.177_{\pm 0.071}$
fluoride	$\mathbf{0.476_{\pm 0.160}}$	$\underline{0.469_{\pm 0.202}}$	$0.241_{\pm 0.078}$	$0.251_{\pm 0.140}$	$0.355_{\pm 0.241}$	$0.104_{\pm 0.087}$	$0.182_{\pm 0.179}$	$0.312_{\pm 0.134}$	$0.151_{\pm 0.089}$

Table 4: Fidelity^{+} scores computed using the Max method (higher is better) with GCN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$\underline{0.236_{\pm 0.224}}$	$0.154_{\pm 0.141}$	$-0.007_{\pm 0.024}$	$0.085_{\pm 0.119}$	$0.002_{\pm 0.027}$	$0.028_{\pm 0.049}$	$0.134_{\pm 0.257}$	$0.010_{\pm 0.039}$	$\mathbf{0.315_{\pm 0.268}}$
alkane	$0.061_{\pm 0.066}$	$0.048_{\pm 0.063}$	$-0.002_{\pm 0.014}$	$0.098_{\pm 0.062}$	$\mathbf{0.187_{\pm 0.137}}$	$0.067_{\pm 0.033}$	$\underline{0.120_{\pm 0.109}}$	$0.009_{\pm 0.014}$	$0.010_{\pm 0.020}$
ames	$\underline{0.177_{\pm 0.047}}$	$0.175_{\pm 0.047}$	$0.033_{\pm 0.017}$	$0.079_{\pm 0.022}$	$0.055_{\pm 0.048}$	$0.051_{\pm 0.020}$	$0.108_{\pm 0.044}$	$0.136_{\pm 0.079}$	$\mathbf{0.193_{\pm 0.066}}$
bace	$\mathbf{0.168_{\pm 0.079}}$	$0.129_{\pm 0.026}$	$0.056_{\pm 0.019}$	$\underline{0.151_{\pm 0.047}}$	$0.096_{\pm 0.039}$	$0.071_{\pm 0.046}$	$0.097_{\pm 0.038}$	$0.136_{\pm 0.082}$	$0.126_{\pm 0.097}$
bbbp	$\mathbf{0.083_{\pm 0.035}}$	$0.052_{\pm 0.013}$	$0.027_{\pm 0.011}$	$0.058_{\pm 0.023}$	$0.017_{\pm 0.011}$	$0.044_{\pm 0.022}$	$0.047_{\pm 0.029}$	$0.026_{\pm 0.029}$	$\underline{0.065_{\pm 0.031}}$
benzene	$\mathbf{0.474_{\pm 0.036}}$	$0.372_{\pm 0.047}$	$0.048_{\pm 0.035}$	$0.149_{\pm 0.078}$	$0.392_{\pm 0.208}$	$0.085_{\pm 0.026}$	$0.322_{\pm 0.212}$	$\underline{0.406_{\pm 0.079}}$	$0.329_{\pm 0.032}$
fluoride	$0.376_{\pm 0.103}$	$\mathbf{0.537_{\pm 0.165}}$	$0.364_{\pm 0.070}$	$0.386_{\pm 0.109}$	$0.407_{\pm 0.155}$	$0.215_{\pm 0.089}$	$\underline{0.408_{\pm 0.208}}$	$0.406_{\pm 0.170}$	$0.228_{\pm 0.075}$

Table 5: Fidelity^{+} scores computed using the Sum method (higher is better) with GCN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.081_{\pm0.080}$	$\mathbf{0.118_{\pm0.083}}$	$0.003_{\pm0.040}$	$0.070_{\pm0.062}$	$0.018_{\pm0.015}$	$0.027_{\pm0.048}$	$0.037_{\pm0.056}$	$0.043_{\pm0.051}$	$0.050_{\pm0.080}$
alkane	$0.024_{\pm0.007}$	$0.457_{\pm0.103}$	$0.150_{\pm0.121}$	$0.304_{\pm0.129}$	$0.221_{\pm0.074}$	$0.131_{\pm0.050}$	$0.262_{\pm0.186}$	$\mathbf{0.804_{\pm0.116}}$	$0.089_{\pm0.062}$
ames	$0.092_{\pm0.027}$	$0.125_{\pm0.068}$	$0.086_{\pm0.074}$	$0.149_{\pm0.071}$	$0.029_{\pm0.018}$	$0.067_{\pm0.077}$	$0.141_{\pm0.075}$	$\mathbf{0.181_{\pm0.125}}$	$0.126_{\pm0.087}$
bace	$0.083_{\pm0.053}$	$\mathbf{0.135_{\pm0.046}}$	$0.053_{\pm0.047}$	$0.100_{\pm0.050}$	$0.039_{\pm0.048}$	$0.036_{\pm0.033}$	$0.053_{\pm0.053}$	$0.045_{\pm0.075}$	$0.025_{\pm0.060}$
bbbp	$0.219_{\pm0.116}$	$\mathbf{0.298_{\pm0.110}}$	$0.134_{\pm0.088}$	$0.159_{\pm0.064}$	$0.121_{\pm0.087}$	$0.122_{\pm0.055}$	$0.136_{\pm0.064}$	$0.148_{\pm0.054}$	$0.157_{\pm0.085}$
benzene	$\mathbf{0.481_{\pm0.026}}$	$0.465_{\pm0.045}$	$0.011_{\pm0.014}$	$0.139_{\pm0.025}$	$0.023_{\pm0.025}$	$0.086_{\pm0.025}$	$0.113_{\pm0.073}$	$0.158_{\pm0.058}$	$0.283_{\pm0.025}$
fluoride	$0.479_{\pm0.097}$	$0.553_{\pm0.144}$	$0.360_{\pm0.142}$	$0.413_{\pm0.155}$	$0.113_{\pm0.221}$	$0.339_{\pm0.053}$	$0.406_{\pm0.101}$	$\mathbf{0.573_{\pm0.170}}$	$0.366_{\pm0.108}$

Table 6: Fidelity$^+$ scores computed using the Mean method (higher is better) with GAT

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.094_{\pm0.089}$	$\mathbf{0.142_{\pm0.110}}$	$-0.014_{\pm0.059}$	$0.038_{\pm0.059}$	$-0.007_{\pm0.056}$	$0.014_{\pm0.052}$	$-0.008_{\pm0.053}$	$0.043_{\pm0.069}$	$0.102_{\pm0.131}$
alkane	$0.041_{\pm0.018}$	$0.241_{\pm0.152}$	$0.165_{\pm0.284}$	$0.196_{\pm0.064}$	$0.053_{\pm0.093}$	$0.061_{\pm0.036}$	$0.211_{\pm0.207}$	$0.321_{\pm0.093}$	$0.091_{\pm0.092}$
ames	$0.102_{\pm0.049}$	$0.091_{\pm0.029}$	$0.084_{\pm0.040}$	$0.107_{\pm0.045}$	$0.020_{\pm0.037}$	$0.062_{\pm0.036}$	$0.108_{\pm0.035}$	$\mathbf{0.143_{\pm0.057}}$	$0.133_{\pm0.029}$
bace	$0.093_{\pm0.036}$	$0.213_{\pm0.045}$	$0.088_{\pm0.023}$	$0.116_{\pm0.050}$	$0.076_{\pm0.066}$	$0.101_{\pm0.038}$	$0.136_{\pm0.094}$	$\mathbf{0.231_{\pm0.103}}$	$0.097_{\pm0.044}$
bbbp	$0.135_{\pm0.091}$	$\mathbf{0.166_{\pm0.130}}$	$0.068_{\pm0.103}$	$0.145_{\pm0.128}$	$0.083_{\pm0.102}$	$0.095_{\pm0.093}$	$0.090_{\pm0.079}$	$0.104_{\pm0.121}$	$0.145_{\pm0.107}$
benzene	$0.472_{\pm0.031}$	$\mathbf{0.512_{\pm0.086}}$	$0.041_{\pm0.023}$	$0.098_{\pm0.032}$	$0.206_{\pm0.184}$	$0.058_{\pm0.015}$	$0.134_{\pm0.068}$	$0.379_{\pm0.048}$	$0.198_{\pm0.070}$
fluoride	$\mathbf{0.718_{\pm0.149}}$	$0.679_{\pm0.169}$	$0.189_{\pm0.211}$	$0.210_{\pm0.081}$	$0.136_{\pm0.071}$	$0.121_{\pm0.022}$	$0.190_{\pm0.140}$	$0.450_{\pm0.263}$	$0.197_{\pm0.079}$

Table 7: Fidelity$^+$ scores computed using the Max method (higher is better) with GAT

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.078_{\pm0.087}$	$\mathbf{0.088_{\pm0.076}}$	$-0.012_{\pm0.040}$	$0.044_{\pm0.074}$	$0.007_{\pm0.010}$	$0.019_{\pm0.046}$	$0.010_{\pm0.052}$	$0.032_{\pm0.045}$	$0.050_{\pm0.164}$
alkane	$0.032_{\pm0.022}$	$0.298_{\pm0.111}$	$0.125_{\pm0.091}$	$0.349_{\pm0.153}$	$0.388_{\pm0.254}$	$0.139_{\pm0.084}$	$0.393_{\pm0.313}$	$\mathbf{0.791_{\pm0.225}}$	$0.125_{\pm0.165}$
ames	$0.114_{\pm0.040}$	$0.120_{\pm0.015}$	$0.042_{\pm0.030}$	$0.106_{\pm0.037}$	$0.046_{\pm0.012}$	$0.070_{\pm0.017}$	$0.109_{\pm0.060}$	$\mathbf{0.177_{\pm0.091}}$	$0.143_{\pm0.045}$
bace	$\mathbf{0.132_{\pm0.037}}$	$0.131_{\pm0.011}$	$0.043_{\pm0.033}$	$0.102_{\pm0.025}$	$0.053_{\pm0.028}$	$0.085_{\pm0.029}$	$0.084_{\pm0.028}$	$0.128_{\pm0.061}$	$0.074_{\pm0.049}$
bbbp	$\mathbf{0.135_{\pm0.060}}$	$0.133_{\pm0.062}$	$0.059_{\pm0.044}$	$0.094_{\pm0.056}$	$0.080_{\pm0.047}$	$0.077_{\pm0.044}$	$0.072_{\pm0.042}$	$0.086_{\pm0.054}$	$0.096_{\pm0.064}$
benzene	$\mathbf{0.483_{\pm0.026}}$	$0.258_{\pm0.073}$	$0.018_{\pm0.008}$	$0.082_{\pm0.030}$	$0.036_{\pm0.026}$	$0.061_{\pm0.022}$	$0.130_{\pm0.056}$	$0.179_{\pm0.054}$	$0.268_{\pm0.046}$
fluoride	$0.520_{\pm0.131}$	$0.564_{\pm0.201}$	$0.341_{\pm0.100}$	$0.382_{\pm0.100}$	$0.093_{\pm0.093}$	$0.261_{\pm0.074}$	$0.290_{\pm0.142}$	$\mathbf{0.616_{\pm0.118}}$	$0.323_{\pm0.099}$

Table 8: Fidelity$^+$ scores computed using the Sum method (higher is better) with GAT

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.127_{\pm 0.184}$	$\underline{0.253_{\pm 0.394}}$	$0.018_{\pm 0.027}$	$0.112_{\pm 0.171}$	$0.015_{\pm 0.023}$	$0.011_{\pm 0.071}$	$0.043_{\pm 0.062}$	$0.043_{\pm 0.069}$	$\mathbf{0.263_{\pm 0.365}}$
alkane	$0.149_{\pm 0.209}$	$\underline{0.222_{\pm 0.301}}$	$0.033_{\pm 0.078}$	$0.002_{\pm 0.220}$	$0.173_{\pm 0.164}$	$0.007_{\pm 0.139}$	$0.221_{\pm 0.274}$	$\mathbf{0.223_{\pm 0.272}}$	$-0.037_{\pm 0.204}$
ames	$0.155_{\pm 0.056}$	$\mathbf{0.258_{\pm 0.092}}$	$0.034_{\pm 0.021}$	$0.124_{\pm 0.042}$	$0.049_{\pm 0.021}$	$0.099_{\pm 0.058}$	$0.137_{\pm 0.101}$	$0.127_{\pm 0.078}$	$\underline{0.189_{\pm 0.062}}$
bace	$0.157_{\pm 0.075}$	$\mathbf{0.276_{\pm 0.052}}$	$0.047_{\pm 0.026}$	$0.175_{\pm 0.062}$	$0.039_{\pm 0.015}$	$0.101_{\pm 0.058}$	$0.163_{\pm 0.099}$	$\underline{0.214_{\pm 0.138}}$	$0.203_{\pm 0.023}$
bbbp	$\mathbf{0.237_{\pm 0.141}}$	$\underline{0.193_{\pm 0.086}}$	$0.111_{\pm 0.076}$	$0.121_{\pm 0.046}$	$0.128_{\pm 0.072}$	$0.093_{\pm 0.034}$	$0.103_{\pm 0.042}$	$0.106_{\pm 0.072}$	$0.127_{\pm 0.076}$
benzene	$0.286_{\pm 0.058}$	$\mathbf{0.422_{\pm 0.053}}$	$0.033_{\pm 0.031}$	$0.166_{\pm 0.047}$	$0.067_{\pm 0.040}$	$0.091_{\pm 0.026}$	$0.221_{\pm 0.139}$	$\underline{0.349_{\pm 0.054}}$	$0.331_{\pm 0.042}$
fluoride	$0.346_{\pm 0.157}$	$\mathbf{0.572_{\pm 0.064}}$	$0.223_{\pm 0.101}$	$0.406_{\pm 0.140}$	$\underline{0.500_{\pm 0.132}}$	$0.312_{\pm 0.031}$	$0.473_{\pm 0.132}$	$0.426_{\pm 0.081}$	$0.306_{\pm 0.037}$

Table 9: Fidelity^{+} scores computed using the Mean method (higher is better) with GIN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.127_{\pm 0.115}$	$\mathbf{0.254_{\pm 0.231}}$	$0.011_{\pm 0.062}$	$0.034_{\pm 0.061}$	$0.009_{\pm 0.056}$	$0.022_{\pm 0.029}$	$0.023_{\pm 0.065}$	$0.013_{\pm 0.043}$	$\underline{0.207_{\pm 0.213}}$
alkane	$\mathbf{0.333_{\pm 0.131}}$	$\underline{0.145_{\pm 0.053}}$	$0.010_{\pm 0.025}$	$0.029_{\pm 0.026}$	$0.020_{\pm 0.025}$	$0.020_{\pm 0.015}$	$0.044_{\pm 0.098}$	$0.044_{\pm 0.065}$	$0.023_{\pm 0.053}$
ames	$0.163_{\pm 0.063}$	$\mathbf{0.222_{\pm 0.064}}$	$0.037_{\pm 0.034}$	$0.109_{\pm 0.047}$	$0.054_{\pm 0.031}$	$0.066_{\pm 0.028}$	$0.086_{\pm 0.067}$	$0.105_{\pm 0.022}$	$\underline{0.177_{\pm 0.043}}$
bace	$\underline{0.128_{\pm 0.093}}$	$\mathbf{0.213_{\pm 0.074}}$	$0.032_{\pm 0.075}$	$0.127_{\pm 0.066}$	$0.051_{\pm 0.028}$	$0.061_{\pm 0.036}$	$0.080_{\pm 0.019}$	$0.086_{\pm 0.044}$	$0.107_{\pm 0.073}$
bbbp	$\mathbf{0.114_{\pm 0.134}}$	$\underline{0.092_{\pm 0.058}}$	$0.033_{\pm 0.045}$	$0.071_{\pm 0.020}$	$0.076_{\pm 0.065}$	$0.052_{\pm 0.024}$	$0.054_{\pm 0.058}$	$0.053_{\pm 0.069}$	$0.069_{\pm 0.053}$
benzene	$\underline{0.485_{\pm 0.021}}$	$\mathbf{0.496_{\pm 0.034}}$	$0.127_{\pm 0.101}$	$0.196_{\pm 0.053}$	$0.220_{\pm 0.174}$	$0.085_{\pm 0.021}$	$0.234_{\pm 0.208}$	$0.434_{\pm 0.083}$	$0.329_{\pm 0.080}$
fluoride	$\mathbf{0.634_{\pm 0.207}}$	$\underline{0.587_{\pm 0.252}}$	$0.060_{\pm 0.029}$	$0.137_{\pm 0.118}$	$0.095_{\pm 0.213}$	$0.088_{\pm 0.050}$	$0.207_{\pm 0.173}$	$0.286_{\pm 0.079}$	$0.153_{\pm 0.119}$

Table 10: Fidelity^{+} scores computed using the Max method (higher is better) with GIN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.165_{\pm 0.233}$	$\mathbf{0.320_{\pm 0.353}}$	$0.002_{\pm 0.012}$	$0.055_{\pm 0.108}$	$0.024_{\pm 0.043}$	$0.010_{\pm 0.032}$	$0.035_{\pm 0.073}$	$0.035_{\pm 0.073}$	$\underline{0.226_{\pm 0.343}}$
alkane	$0.052_{\pm 0.068}$	$0.059_{\pm 0.045}$	$0.013_{\pm 0.018}$	$\underline{0.127_{\pm 0.057}}$	$\mathbf{0.210_{\pm 0.103}}$	$0.082_{\pm 0.033}$	$0.071_{\pm 0.093}$	$0.102_{\pm 0.084}$	$0.051_{\pm 0.051}$
ames	$0.191_{\pm 0.118}$	$\mathbf{0.296_{\pm 0.108}}$	$0.042_{\pm 0.027}$	$0.173_{\pm 0.053}$	$0.080_{\pm 0.026}$	$0.129_{\pm 0.067}$	$0.142_{\pm 0.080}$	$0.133_{\pm 0.063}$	$\underline{0.222_{\pm 0.114}}$
bace	$0.148_{\pm 0.050}$	$\mathbf{0.244_{\pm 0.107}}$	$0.040_{\pm 0.037}$	$0.152_{\pm 0.059}$	$0.053_{\pm 0.032}$	$0.095_{\pm 0.055}$	$0.129_{\pm 0.074}$	$0.181_{\pm 0.087}$	$\underline{0.184_{\pm 0.069}}$
bbbp	$\mathbf{0.159_{\pm 0.069}}$	$0.080_{\pm 0.031}$	$0.042_{\pm 0.026}$	$0.057_{\pm 0.016}$	$0.039_{\pm 0.016}$	$0.055_{\pm 0.032}$	$0.076_{\pm 0.024}$	$0.077_{\pm 0.034}$	$\underline{0.090_{\pm 0.047}}$
benzene	$\mathbf{0.498_{\pm 0.044}}$	$\underline{0.491_{\pm 0.080}}$	$0.016_{\pm 0.017}$	$0.117_{\pm 0.040}$	$0.113_{\pm 0.031}$	$0.078_{\pm 0.019}$	$0.162_{\pm 0.153}$	$0.384_{\pm 0.110}$	$0.336_{\pm 0.076}$
fluoride	$0.309_{\pm 0.109}$	$\mathbf{0.467_{\pm 0.261}}$	$0.143_{\pm 0.097}$	$0.313_{\pm 0.212}$	$0.301_{\pm 0.204}$	$0.192_{\pm 0.145}$	$\underline{0.358_{\pm 0.194}}$	$0.353_{\pm 0.276}$	$0.204_{\pm 0.188}$

Table 11: Fidelity^{+} scores computed using the Sum method (higher is better) with GIN

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.222_{\pm 0.168}$	$\mathbf{0.288_{\pm 0.212}}$	$0.005_{\pm 0.022}$	$0.125_{\pm 0.108}$	$0.061_{\pm 0.075}$	$0.011_{\pm 0.031}$	$0.065_{\pm 0.059}$	$0.191_{\pm 0.221}$	$\underline{0.265_{\pm 0.207}}$
alkane	$0.104_{\pm 0.071}$	$\underline{0.432_{\pm 0.139}}$	$0.093_{\pm 0.051}$	$0.271_{\pm 0.057}$	$0.427_{\pm 0.272}$	$0.168_{\pm 0.021}$	$0.218_{\pm 0.108}$	$\mathbf{0.564_{\pm 0.057}}$	$0.149_{\pm 0.075}$
ames	$0.147_{\pm 0.020}$	$\underline{0.182_{\pm 0.042}}$	$0.083_{\pm 0.039}$	$0.169_{\pm 0.036}$	$0.070_{\pm 0.043}$	$0.083_{\pm 0.026}$	$0.104_{\pm 0.055}$	$\mathbf{0.211_{\pm 0.052}}$	$0.160_{\pm 0.061}$
bace	$0.145_{\pm 0.046}$	$0.155_{\pm 0.055}$	$0.074_{\pm 0.038}$	$\underline{0.239_{\pm 0.120}}$	$0.094_{\pm 0.032}$	$0.137_{\pm 0.044}$	$0.177_{\pm 0.099}$	$\mathbf{0.240_{\pm 0.103}}$	$0.110_{\pm 0.037}$
bbbp	$\underline{0.152_{\pm 0.065}}$	$\mathbf{0.192_{\pm 0.112}}$	$0.037_{\pm 0.028}$	$0.072_{\pm 0.031}$	$0.056_{\pm 0.045}$	$0.052_{\pm 0.024}$	$0.081_{\pm 0.059}$	$0.054_{\pm 0.022}$	$0.076_{\pm 0.030}$
benzene	$0.417_{\pm 0.071}$	$\mathbf{0.502_{\pm 0.050}}$	$0.015_{\pm 0.055}$	$0.130_{\pm 0.046}$	$0.195_{\pm 0.171}$	$0.081_{\pm 0.026}$	$0.132_{\pm 0.135}$	$\underline{0.424_{\pm 0.145}}$	$0.324_{\pm 0.049}$
fluoride	$0.348_{\pm 0.159}$	$\mathbf{0.564_{\pm 0.134}}$	$0.468_{\pm 0.204}$	$0.435_{\pm 0.098}$	$0.398_{\pm 0.219}$	$0.278_{\pm 0.061}$	$0.480_{\pm 0.229}$	$\underline{0.485_{\pm 0.133}}$	$0.358_{\pm 0.160}$

Table 12: Fidelity$^+$ scores computed using the Mean method (higher is better) with GraphSAGE

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$0.161_{\pm 0.128}$	$\underline{0.232_{\pm 0.142}}$	$-0.003_{\pm 0.095}$	$0.030_{\pm 0.050}$	$-0.016_{\pm 0.100}$	$0.007_{\pm 0.100}$	$-0.000_{\pm 0.093}$	$0.125_{\pm 0.092}$	$\mathbf{0.243_{\pm 0.174}}$
alkane	$\underline{0.497_{\pm 0.062}}$	$\mathbf{0.548_{\pm 0.102}}$	$0.127_{\pm 0.205}$	$0.058_{\pm 0.025}$	$0.439_{\pm 0.359}$	$0.022_{\pm 0.049}$	$0.116_{\pm 0.165}$	$0.062_{\pm 0.125}$	$0.022_{\pm 0.036}$
ames	$\mathbf{0.155_{\pm 0.031}}$	$\underline{0.138_{\pm 0.078}}$	$0.051_{\pm 0.049}$	$0.100_{\pm 0.016}$	$0.057_{\pm 0.018}$	$0.049_{\pm 0.018}$	$0.079_{\pm 0.046}$	$0.120_{\pm 0.033}$	$0.105_{\pm 0.039}$
bace	$0.151_{\pm 0.063}$	$\mathbf{0.214_{\pm 0.070}}$	$0.092_{\pm 0.064}$	$0.141_{\pm 0.053}$	$0.131_{\pm 0.048}$	$0.072_{\pm 0.028}$	$0.108_{\pm 0.038}$	$\underline{0.163_{\pm 0.088}}$	$0.104_{\pm 0.046}$
bbbp	$\underline{0.049_{\pm 0.020}}$	$0.037_{\pm 0.021}$	$-0.001_{\pm 0.010}$	$\mathbf{0.059_{\pm 0.017}}$	$0.015_{\pm 0.024}$	$0.034_{\pm 0.017}$	$0.041_{\pm 0.045}$	$0.012_{\pm 0.014}$	$0.045_{\pm 0.032}$
benzene	$0.492_{\pm 0.044}$	$\underline{0.636_{\pm 0.102}}$	$0.112_{\pm 0.079}$	$0.331_{\pm 0.118}$	$\mathbf{0.650_{\pm 0.084}}$	$0.120_{\pm 0.063}$	$0.367_{\pm 0.270}$	$0.498_{\pm 0.077}$	$0.349_{\pm 0.055}$
fluoride	$\mathbf{0.521_{\pm 0.165}}$	$\underline{0.401_{\pm 0.111}}$	$0.220_{\pm 0.101}$	$0.185_{\pm 0.091}$	$0.154_{\pm 0.184}$	$0.096_{\pm 0.052}$	$0.325_{\pm 0.125}$	$0.141_{\pm 0.045}$	$0.187_{\pm 0.072}$

Table 13: Fidelity$^+$ scores computed using the Max method (higher is better) with GraphSAGE

Dataset	Group Lasso	Lasso	PGExp	GNNExp	Grad-CAM	Graph Mask	GBP	SA	SubgraphX
MUTAG	$\underline{0.194_{\pm 0.102}}$	$0.192_{\pm 0.091}$	$0.014_{\pm 0.013}$	$0.104_{\pm 0.092}$	$0.023_{\pm 0.024}$	$0.041_{\pm 0.038}$	$0.026_{\pm 0.037}$	$0.066_{\pm 0.074}$	$\mathbf{0.364_{\pm 0.284}}$
alkane	$0.129_{\pm 0.108}$	$0.035_{\pm 0.032}$	$0.028_{\pm 0.027}$	$0.137_{\pm 0.037}$	$\mathbf{0.557_{\pm 0.231}}$	$0.069_{\pm 0.053}$	$0.115_{\pm 0.069}$	$\underline{0.175_{\pm 0.083}}$	$0.077_{\pm 0.055}$
ames	$0.137_{\pm 0.033}$	$\underline{0.163_{\pm 0.023}}$	$0.082_{\pm 0.045}$	$0.119_{\pm 0.024}$	$0.068_{\pm 0.026}$	$0.079_{\pm 0.016}$	$0.101_{\pm 0.040}$	$\mathbf{0.174_{\pm 0.059}}$	$0.135_{\pm 0.059}$
bace	$0.173_{\pm 0.084}$	$\underline{0.185_{\pm 0.075}}$	$0.130_{\pm 0.057}$	$0.178_{\pm 0.072}$	$0.056_{\pm 0.055}$	$0.154_{\pm 0.093}$	$0.182_{\pm 0.067}$	$\mathbf{0.207_{\pm 0.106}}$	$0.172_{\pm 0.091}$
bbbp	$\underline{0.059_{\pm 0.039}}$	$\mathbf{0.066_{\pm 0.037}}$	$0.013_{\pm 0.007}$	$0.045_{\pm 0.018}$	$0.020_{\pm 0.007}$	$0.027_{\pm 0.017}$	$0.041_{\pm 0.029}$	$0.026_{\pm 0.011}$	$0.028_{\pm 0.009}$
benzene	$\mathbf{0.400_{\pm 0.066}}$	$0.382_{\pm 0.057}$	$0.062_{\pm 0.040}$	$0.133_{\pm 0.065}$	$0.172_{\pm 0.140}$	$0.083_{\pm 0.039}$	$0.163_{\pm 0.150}$	$\underline{0.385_{\pm 0.085}}$	$0.327_{\pm 0.062}$
fluoride	$\mathbf{0.293_{\pm 0.059}}$	$0.275_{\pm 0.142}$	$0.168_{\pm 0.163}$	$0.219_{\pm 0.104}$	$0.124_{\pm 0.068}$	$0.154_{\pm 0.060}$	$0.257_{\pm 0.139}$	$\underline{0.284_{\pm 0.075}}$	$0.231_{\pm 0.107}$

Table 14: Fidelity$^+$ scores computed using the Sum method (higher is better) with GraphSAGE